EXPERIMENT - 01

Library Management System

Aim:

To design and implement a Web-based Library Management System that allows students and librarian administrators to efficiently manage catalog browsing, book issue/return tracking, fine calculations, and semester-wise study material downloads.

Procedure:

1. Analyze requirements for both Librarian Admin and Student users to identify essential features such as authentication, book cataloging, search functionality, fine status, and material downloads.

2. Design the front-end user interface using HTML structure, embedded styles, and interactive elements for smooth navigation across different modules.

3. Implement the Librarian Admin module with capabilities to handle inventory management, issue books, update return statuses, apply overdue fines, and manage study materials by academic year and semester.

4. Implement the Student module allowing registered users to search catalog items, review borrow history, track issue/expiry dates, check pending fines, and download course-specific material.

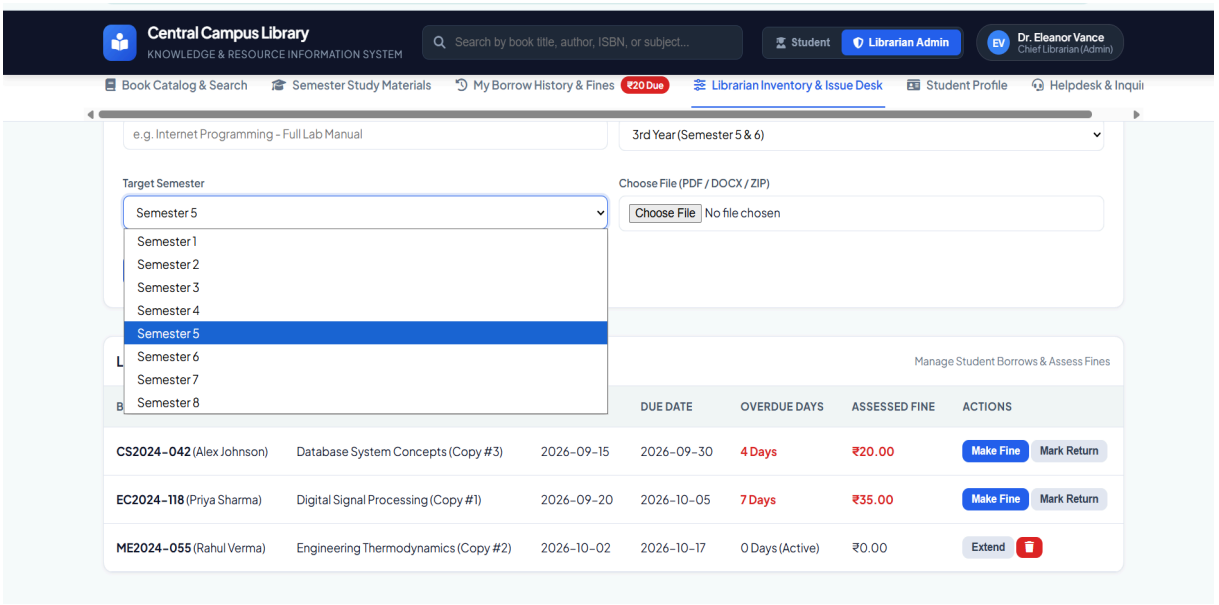
5. Test and verify system workflows by simulating user logins, searching for catalog resources, reviewing borrow records, and checking interface responses.

Code:

```
index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Central University Library &amp;amp; Resource Information System</title>
  <!-- Google Fonts -->
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700;800&family=Playfair+Display:wght@600;700&disp
  <!-- Lucide Icons (CDN) -->
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css">

  <style>
    :root {
      --primary: #1e3a8a;
      --primary-light: #2563eb;
      --primary-subtle: #dbeafe;
      --accent: #d97706;
      --accent-soft: #fef3c7;
      --success: #059669;
      --success-soft: #d1fae5;
      --danger: #dc2626;
      --danger-soft: #fee2e2;
      --bg-main: #f8fafc;
      --card-bg: #ffffff;
      --text-main: #0f172a;
      --text-muted: #64748b;
      --border: #e2e8f0;
      --radius: 12px;
```

Output:



Result:

The Library Management System web interface was successfully designed, developed, and verified with all specified features for students and library administration.

Experiment – 2

Online Hotel Room Booking System

Aim:

To design and implement a web-based online hotel room booking system that enables users to search for hotels, check room availability, manage bookings, and complete secure payments, while providing administrators with dedicated controls to manage hotel listings, room prices, availability, and static site content.

Procedure:

1. Analyze functional requirements for both registered users and administrators to define core capabilities including user authentication, hotel search filters, reservation management, policy-driven cancellation/rescheduling, and admin-side content management.

2. Design a responsive user interface with structured HTML, CSS custom properties, navigation bars, search filters, and interactive reservation forms ensuring seamless navigation on all screen sizes.

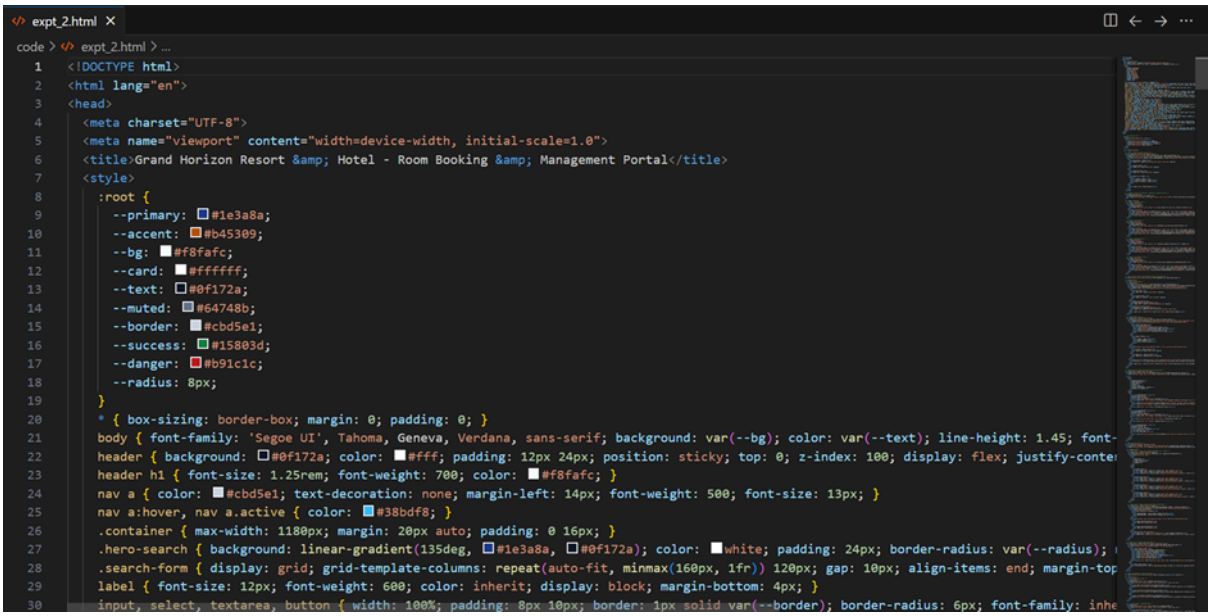
3. Build user dashboard features allowing guests to search available rooms, select stay dates, manage personal profile details, review booking history, check-in statuses, and request date changes or cancellations up to 3 days prior to arrival.

4. Implement the administrator control panel allowing admins to adjust room quantities and pricing across all room types, manage existing customer bookings, process cancellations, and update static content such as room descriptions, FAQs, local travel guides, and privacy policies.

5. Integrate secure payment workflows and post-booking functionalities including automated confirmation emails, review submissions, and rating features.

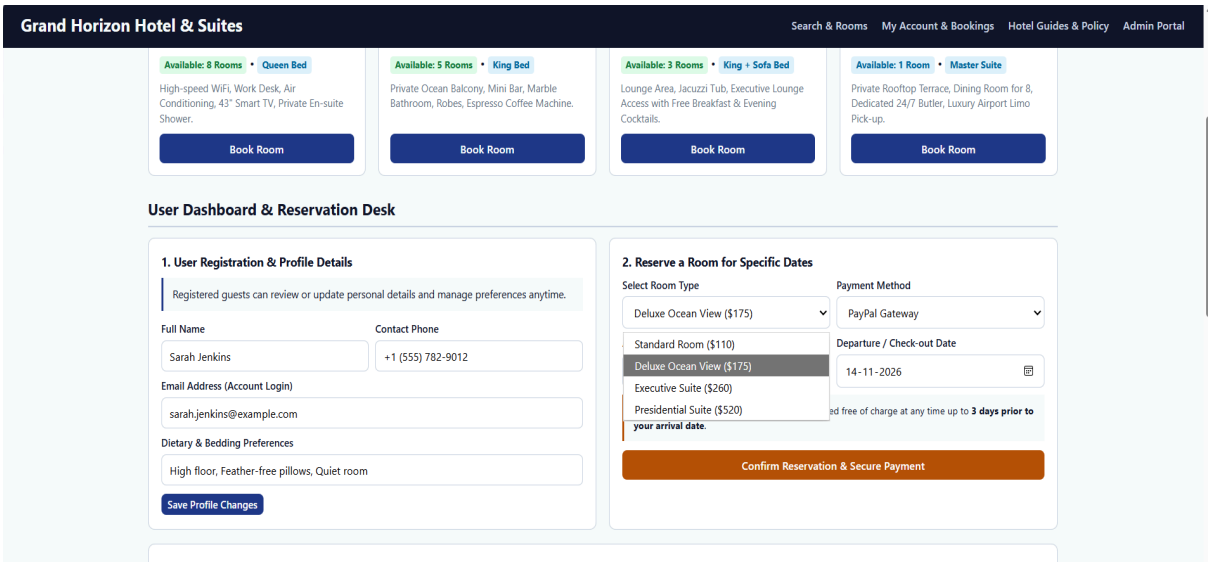
6. Test and verify all system workflows by performing end-to-end user registration, room availability lookup, reservation creation, booking modifications, admin updates, and interface responsiveness checks.

Code:



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Grand Horizon Resort &amp;amp; Hotel - Room Booking &amp;amp; Management Portal</title>
  <style>
    :root {
      --primary: #1e3a8a;
      --accent: #b45309;
      --bg: #f8fafc;
      --card: #ffffff;
      --text: #0f172a;
      --muted: #64748b;
      --border: #cbd5e1;
      --success: #15803d;
      --danger: #b91c1c;
      --radius: 8px;
    }
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background: var(--bg); color: var(--text); line-height: 1.45; font-
    header { background: #0f172a; color: #fff; padding: 12px 24px; position: sticky; top: 0; z-index: 100; display: flex; justify-conte
    header h1 { font-size: 1.25rem; font-weight: 700; color: #f8fafc; }
    nav a { color: #cbd5e1; text-decoration: none; margin-left: 14px; font-weight: 500; font-size: 13px; }
    nav a:hover, nav a.active { color: #38bdf8; }
    .container { max-width: 1180px; margin: 20px auto; padding: 0 16px; }
    .hero-search { background: linear-gradient(135deg, #1e3a8a, #0f172a); color: white; padding: 24px; border-radius: var(--radius);
    .search-form { display: grid; grid-template-columns: repeat(auto-fit, minmax(160px, 1fr)) 120px; gap: 10px; align-items: end; margin-top
    label { font-size: 12px; font-weight: 600; color: inherit; display: block; margin-bottom: 4px; }
    input, select, textarea, button { width: 100%; padding: 8px 10px; border: 1px solid var(--border); border-radius: 6px; font-family: inhe
```

Output:



Result:

The online hotel room booking web application was successfully designed, developed, and verified with all specified functionalities for guests and administrators.

EXPERIMENT - 03

Online Bus Ticket Booking System

Aim:

To design and implement a web-based online bus ticket booking system that enables users to search for buses by route, date, and time, select available seats, manage bookings via a personal dashboard, and complete secure payments, while enabling administrators to manage bus schedules, categories, seat availability, news, user accounts, and system reports.

Procedure:

1. Analyze system requirements for both registered passengers and system administrators to define key features including authentication, bus cataloging, seat selection, booking cancellations, user feedback, and report generation.

2. Design a mobile-responsive user interface with clear layout structure, navigation controls, search forms, and interactive seat selection options.

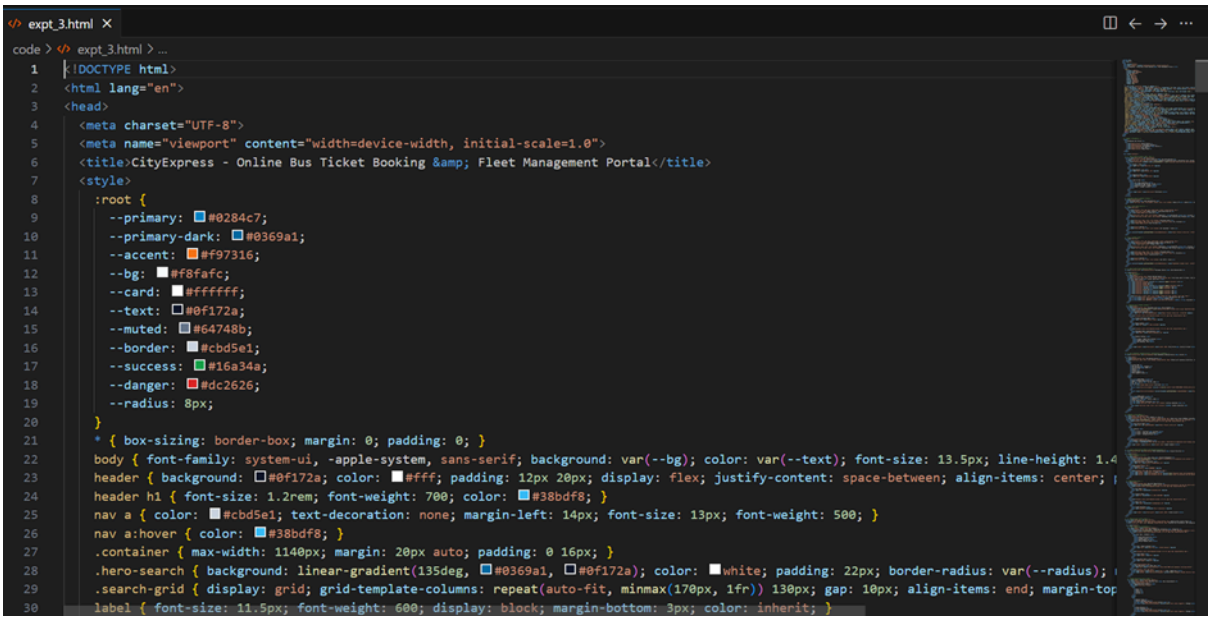
3. Implement authentication and registration modules allowing users to create accounts, log in, perform password management, and access personal dashboards.

4. Implement ticket reservation and management workflows including route lookup, seat selection, passenger detail input, secure payment processing, and cancellation capabilities.

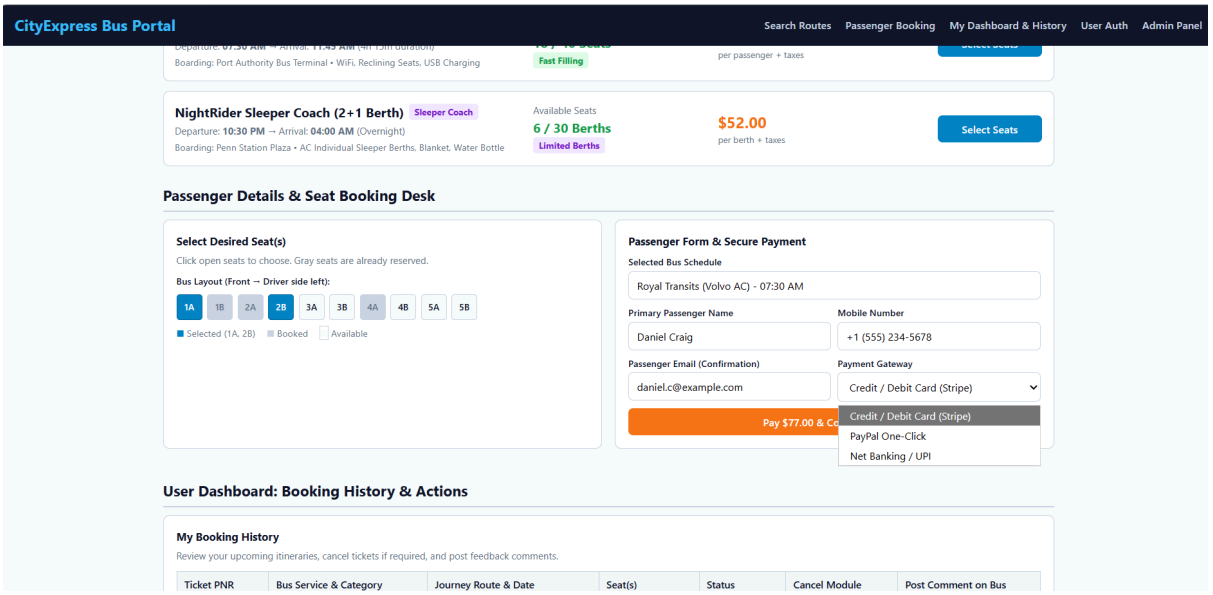
5. Build the administrator control module to manage bus details (type, photo, description, location, available seats), handle bus categories (Volvo, Sleeper coach, etc.), monitor registered users, manage site news/comments, and generate administrative reports.

6. Test and verify system functionality by executing complete user and admin workflows, verifying booking confirmation emails, testing comment submissions, and checking interface responsiveness across devices.

Code:



Output:



Result:

The online bus ticket booking web application was successfully designed, developed, and verified with all specified functionalities for passengers and administrators.

EXPERIMENT - 04

Online Train Reservation System

Aim:

To design and implement a web-based online train reservation system that enables passengers to search for trains by route, date, and class, check seat availability, view PNR status, manage reservations, and process cancellations and refunds, while providing administrators with dedicated controls to manage train schedules, fare structures, and seat allocations.

Procedure:

1. Analyze system requirements for passengers and administrators to identify core functionalities including user login, train route search, class selection, ticket booking, PNR generation, cancellation processing, and admin maintenance.

2. Design a responsive user interface with clean navigation, interactive search forms, passenger detail inputs, and status lookup sections.

3. Implement user authentication allowing passengers to log in with valid credentials, select journey dates, origin/destination stations, and ticket class types.

4. Integrate ticket booking and payment workflows to display available train lists, process payments, generate unique PNR numbers, and issue digital tickets.

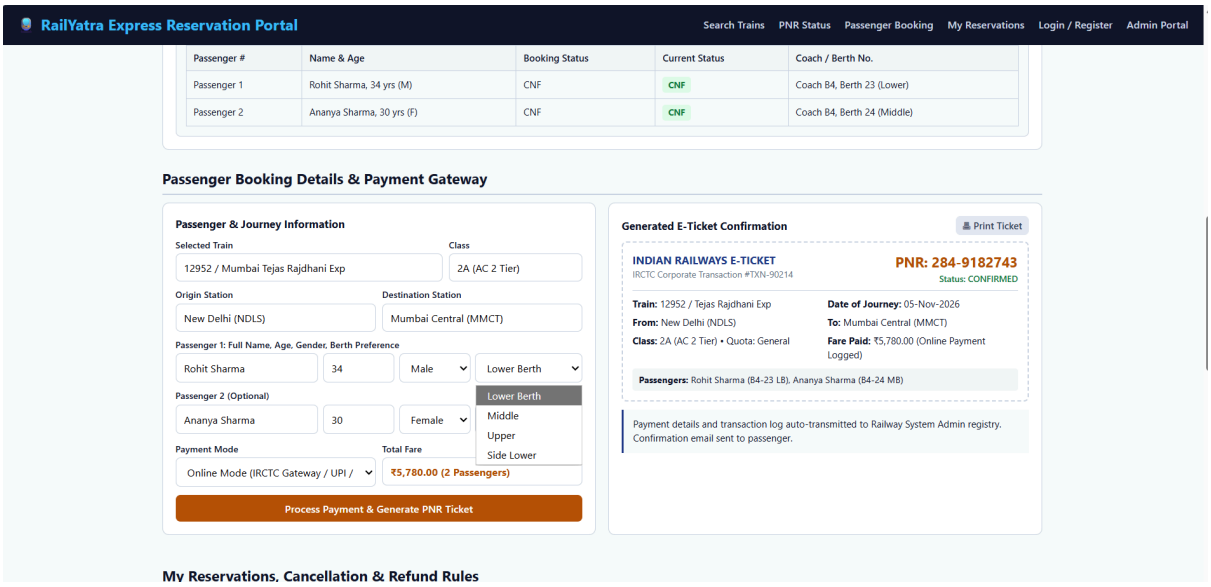
5. Build cancellation and refund workflows adhering to policy-based percentage calculations based on cancellation dates and ticket status.

6. Implement the administrator panel for managing train details, zone-wise fare charts, seat inventory, and viewing payment records.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>RailYatra - Indian Railways Online Train Reservation Portal</title>
  <style>
    :root {
      --primary: #1e3a8a;
      --primary-dark: #0f172a;
      --accent: #b45309;
      --bg: #f8fafc;
      --card: #ffffff;
      --text: #0f172a;
      --muted: #64748b;
      --border: #cbd5e1;
      --success: #15803d;
      --warning: #d97706;
      --danger: #b91c1c;
      --radius: 8px;
    }
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body { font-family: system-ui, -apple-system, sans-serif; background: var(--bg); color: var(--text); font-size: 13.5px; line-height: 1.4
    header { background: #0f172a; color: #fff; padding: 12px 20px; display: flex; justify-content: space-between; align-items: center;
    header h1 { font-size: 1.2rem; font-weight: 700; color: #38bdf8; display: flex; align-items: center; gap: 8px; }
    nav a { color: #cbd5e1; text-decoration: none; margin-left: 14px; font-size: 13px; font-weight: 500; }
    nav a:hover { color: #38bdf8; }
    .container { max-width: 1140px; margin: 20px auto; padding: 0 16px; }
    .hero-search { background: linear-gradient(135deg, #1e3a8a, #0f172a); color: white; padding: 22px; border-radius: var(--radius);
    .search-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(170px, 1fr)) 130px; gap: 10px; align-items: end; margin-top
```

Output:



Result:

The online train reservation system web application was successfully designed, developed, and verified with all specified functionalities for passengers and administrators.

EXPERIMENT - 05

Online Flight Reservation System

Aim:

To design and implement a web-based online flight reservation system that consolidates real-time flight schedules, seat availability, and fares across airlines, enabling users to search, book, and manage e-tickets, while providing administrators with tools to control flight inventory and user accounts.

Procedure:

1. Analyze functional requirements for customers, travel agents, and administrators to define key modules including user login, flight search, ticket booking, payment processing, e-ticket printing, and cancellation handling.

2. Design responsive front-end interfaces for essential user pages including Administrator User Login, Forgot Password, Search Flight, Book Ticket, Ticket Payment, My Ticket, Cancel Ticket, Customer Feedback, Admin Report, View Booking Status, and Print Ticket.

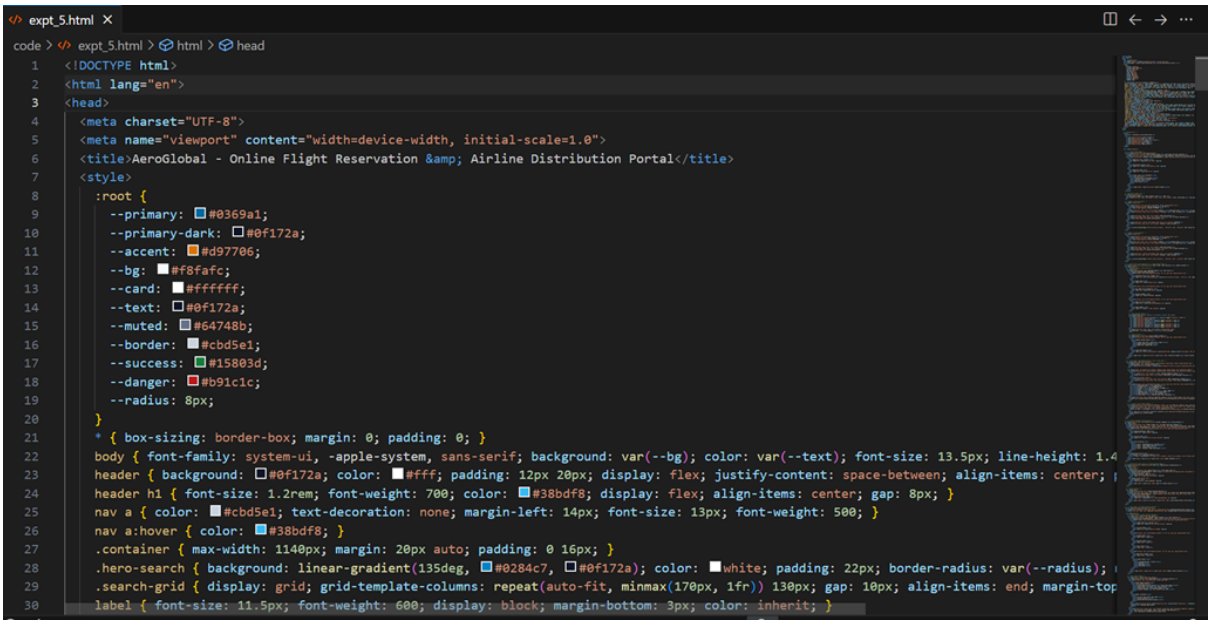
3. Build customer account management workflows allowing registered users to search flights by destination, travel date, and cabin class with real-time seat availability updates.

4. Implement reservation and payment gateway integration to process flight bookings, issue confirmation emails with downloadable e-tickets, and store travel itineraries.

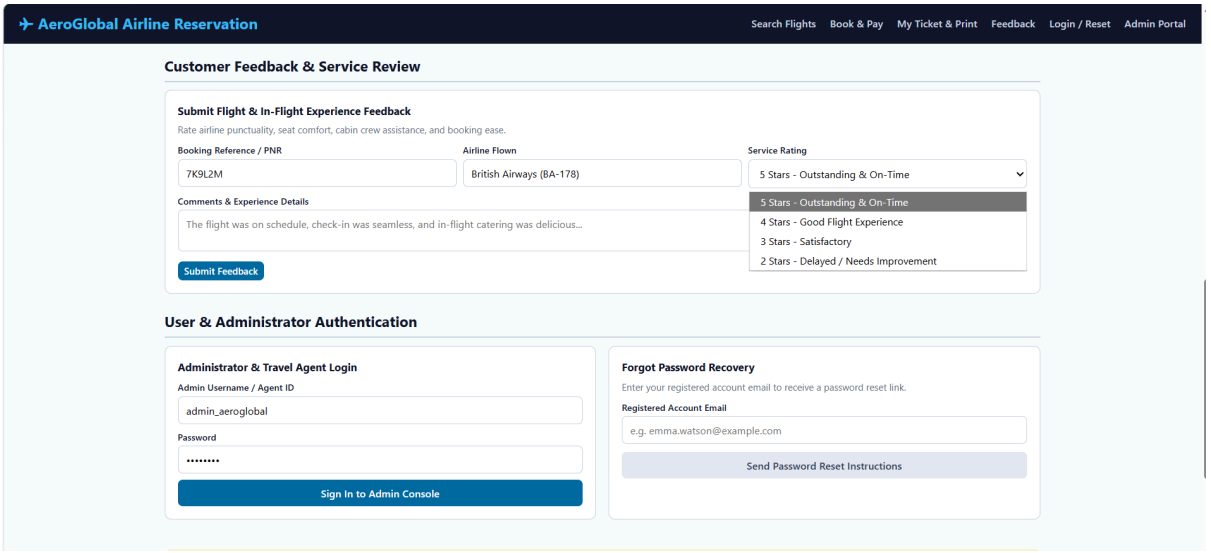
5. Develop administrator management controls for maintaining flight data, fare tiers, schedule adjustments, user accounts, and generating system reports.

6. Test and verify complete system operations by simulating end-to-end flight searches, ticket purchases, itinerary printing, cancellations, and responsive layout checks across devices.

Code:



Output:



Result:

The online flight reservation web application was successfully designed, developed, and verified with all specified functionalities for passengers and administrators.